



DECAY AND RECYCLING

When organisms die and decay, they are recycled by decomposers—fungi and bacteria. These break organisms down into nutrients—substances that can feed living plants. The decomposers are a vital part of any ecosystem, because without them the plants would run out of nutrients. This fallen apple is being decomposed by fungi.



FOOD PYRAMIDS

Any ecosystem has many producers, such as oceanic phytoplankton, fewer consumers, such as zooplankton (microscopic animals), and even fewer secondary consumers, such as fish. This is because some of the energy at each level is turned into activity rather than food. It takes a vast weight of phytoplankton to support one polar bear.

Zooplankton feed on countless phytoplankton

Phytoplankton make their own food



SUCCESSION

Most ecosystems become more complex as they develop over time—a process known as succession. This lone plant growing on the scorched slope of a volcano may be pioneering the way for a more complex ecosystem.



WORLD BIOMES

The world is divided into several large ecosystems called biomes. These include the oceans and vast swaths of landscape such as deserts, grasslands, and forests. Each biome has its own special features, which in most cases are shaped by the climate. Together, these biomes make up the biosphere—the ecosystem of the entire planet.

